

Qudit Stabilizer Simulation Reference

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Abstract—A self-contained reference for tableau-based simulation of Clifford circuits on qudits of prime or composite dimension d . Covers the stabilizer representation, gate conjugation rules, and a measurement protocol using Smith Normal Form reduction.

Index Terms—Clifford Circuit, Smith Normal Form, Tableau Representation, Qudit

I. REPRESENTATION

A Pauli operator on n qudits of dimension d is defined as

$$P = \tau^{-\varphi} Z^{\mathbf{z}} X^{\mathbf{x}}, \quad \mathbf{z}, \mathbf{x} \in \mathbb{Z}_d^n, \quad \varphi \in \mathbb{Z}_D \quad (1)$$

where $\omega = e^{2\pi i/d}$, $\tau = e^{i\pi(d^2+1)/d}$, $\tau^2 = \omega$, and

$$D = \begin{cases} d & d \text{ odd} \\ 2d & d \text{ even.} \end{cases} \quad (2)$$

Two Paulis thus satisfy the commutation relation

$$P_1 P_2 = \tau^{2\langle 1,2 \rangle} P_2 P_1, \quad (3)$$

where $\langle 1,2 \rangle$ is shorthand for the symplectic inner product of the Pauli vectors of P_1 and P_2 :

$$\begin{aligned} \langle 1,2 \rangle &:= \langle \mathbf{z}_1 \oplus \mathbf{x}_1, \mathbf{z}_2 \oplus \mathbf{x}_2 \rangle \\ &= \mathbf{z}_1 \cdot \mathbf{x}_2 - \mathbf{x}_1 \cdot \mathbf{z}_2 \pmod{d} \end{aligned} \quad (4)$$

with $\oplus$ denoting the direct sum of vectors and $(\cdot)$ an inner product. Thus the Paulis commute iff $\langle 1,2 \rangle = 0$. Throughout, two identities will be used. First the *ordered product rule*:

$$\begin{aligned} &(\tau^{-\varphi_1} Z^{\mathbf{z}_1} X^{\mathbf{x}_1})(\tau^{-\varphi_2} Z^{\mathbf{z}_2} X^{\mathbf{x}_2}) \\ &= \tau^{-(\varphi_1 + \varphi_2 + 2\mathbf{x}_1 \cdot \mathbf{z}_2)} Z^{\mathbf{z}_1 + \mathbf{z}_2} X^{\mathbf{x}_1 + \mathbf{x}_2} \end{aligned} \quad (5)$$

and the *power rule*:

$$(\tau^{-\varphi} Z^{\mathbf{z}} X^{\mathbf{x}})^c = \tau^{-(c\varphi + c(c-1)\mathbf{z} \cdot \mathbf{x})} Z^{c\mathbf{z}} X^{c\mathbf{x}}. \quad (6)$$

The state is encoded in a column-major tableau with ZX ordering shown in Fig. 1. Each column is one stabilizer generator $S_j = \tau^{-\varphi_j} Z^{\mathbf{z}_j} X^{\mathbf{x}_j}$. Equivalently,

$$T \in \mathbb{Z}_D^{1 \times \ell} \oplus \mathbb{Z}_d^{n \times \ell} \oplus \mathbb{Z}_d^{n \times \ell}, \quad \ell \leq 2n, \quad (7)$$

Writing those blocks explicitly:

$$\text{phase} \left[\begin{array}{c} \varphi_1 \cdots \varphi_\ell \\ \hline \begin{array}{c} z_{1,1} \cdots z_{1,\ell} \\ \vdots \\ z_{n,1} \cdots z_{n,\ell} \end{array} \\ \hline \begin{array}{c} x_{1,1} \cdots x_{1,\ell} \\ \vdots \\ x_{n,1} \cdots x_{n,\ell} \end{array} \end{array} \right] \in \mathbb{Z}_D^{1 \times \ell} \left. \begin{array}{l} \left. \begin{array}{c} \left. \begin{array}{c} z_{1,1} \cdots z_{1,\ell} \\ \vdots \\ z_{n,1} \cdots z_{n,\ell} \end{array} \right\} n \text{ rows} \in \mathbb{Z}_d^{n \times \ell} \\ \left. \begin{array}{c} x_{1,1} \cdots x_{1,\ell} \\ \vdots \\ x_{n,1} \cdots x_{n,\ell} \end{array} \right\} n \text{ rows} \in \mathbb{Z}_d^{n \times \ell} \end{array} \right\} \end{array} \right.$$

Fig. 1: Stabilizer tableau layout.

For the initial state $|0\rangle^{\otimes n}$ with $\ell = n$, we set the tableau to have the following form:

$$T : \quad \varphi = \mathbf{0}, \quad \mathbf{Z} = \mathbf{1}_n, \quad \mathbf{X} = \mathbf{0}_n \quad (8)$$

For composite d , the number of generators ℓ can grow beyond n (up to $2n$) after measurements. For prime d , $\ell = n$ always.

A. Generator Arithmetic

Two column-level operations are needed. To *multiply* generator k by c copies of generator p (implementing $S_k \leftarrow S_p^c S_k$), apply

$$\begin{aligned} \varphi_k &\leftarrow c\varphi_p + \varphi_k + 2c(\mathbf{x}_p \cdot \mathbf{z}_k) \\ &\quad + c(c-1)(\mathbf{z}_p \cdot \mathbf{x}_p) \end{aligned} \quad (9)$$

$$\mathbf{z}_k \leftarrow c\mathbf{z}_p + \mathbf{z}_k, \quad \mathbf{x}_k \leftarrow c\mathbf{x}_p + \mathbf{x}_k \quad (10)$$

where $\mathbf{z}_k$ in the cross term is the value *before* update.

To *scale* generator p by c (implementing $S_p \leftarrow S_p^c$), the power rule gives

$$\varphi_p \leftarrow c\varphi_p + c(c-1)(\mathbf{z}_p \cdot \mathbf{x}_p) \quad (11)$$

$$\mathbf{z}_p \leftarrow c\mathbf{z}_p, \quad \mathbf{x}_p \leftarrow c\mathbf{x}_p. \quad (12)$$

B. Unimodular Column Transformation

When we need to change basis among the generators (e.g. during measurement), we apply a unimodular matrix V to the tableau columns. Column j of the result represents the ordered product $\prod_k S_k^{V_{kj}}$. The Z and X blocks transform linearly:

$$\mathbf{Z}' = \mathbf{Z}V, \quad \mathbf{X}' = \mathbf{X}V \quad (13)$$

and the phases pick up cross terms from the ordering:

$$\begin{aligned} \varphi'_j &= \sum_k V_{kj} \varphi_k + \sum_k V_{kj} (V_{kj} - 1) G_{kk} \\ &\quad + 2 \sum_{m < k} V_{mj} V_{kj} G_{km} \end{aligned} \quad (14)$$

where $G_{ab} = \mathbf{z}_a \cdot \mathbf{x}_b$ is the Gram matrix of the original (pre-transformation) generators.

II. GATE CONJUGATION

Each Clifford gate is applied by conjugating every column of the tableau: $S_j \leftarrow U S_j U^\dagger$. All phase corrections below use Z/X values *before* the update.

TABLE I: CLIFFORD GATE CONJUGATION RULES.

Gate	Z/X Update	$\Delta\varphi$
S_q	$z_q \leftarrow z_q + x_q$	Refer to (16)
F_q	$(z_q, x_q) \leftarrow (x_q, -z_q)$	$-2z_q x_q$
$F_q^\dagger$	$(z_q, x_q) \leftarrow (-x_q, z_q)$	$-2z_q x_q$
$M_{a,q}$	$z_q \leftarrow a^{-1} z_q, x_q \leftarrow a x_q$	0
$\text{SUM}_{c \rightarrow t}$	$x_t \leftarrow x_t + x_c, z_c \leftarrow z_c - z_t$	0
$\text{CZ}_{1,2}$	$z_1 \leftarrow z_1 + x_2, z_2 \leftarrow z_2 + x_1$	$2x_1 x_2$
X_q	—	$+2z_q$
Z_q	—	$-2x_q$
SWAP_{q_1, q_2}	swap rows $q_1 \leftrightarrow q_2$	0

The Fourier gate F_q swaps the Z and X roles (with a sign flip and phase correction), the multiplier gate $M_{a,q}$ rescales by $a \in \mathbb{Z}_d^\times$, and the two-qudit gates SUM and CZ act on their respective control/target pairs.

The phase gate on qudit q is defined as

$$S = \begin{cases} \sum_{q=0}^{d-1} \tau^{q^2} |q\rangle\langle q| & d \text{ even} \\ \sum_{q=0}^{d-1} \omega^{q(q-1)/2} |q\rangle\langle q| & d \text{ odd.} \end{cases} \quad (15)$$

Both forms satisfy $SZS^\dagger = Z$ (Z is unchanged) and map $X \mapsto ZX$ up to a phase absorbed into the definition. Per column:

$$z_q \leftarrow z_q + x_q, \quad \Delta\varphi = \begin{cases} x_q^2 & d \text{ even} \\ x_q(x_q + 1) & d \text{ odd.} \end{cases} \quad (16)$$

III. MEASUREMENT

We want to measure a Pauli operator $P = \tau^{-\delta} Z^a X^b$. For a computational-basis measurement of qudit r , this is just:

$$Z_r : \text{set } \mathbf{a} = \hat{e}_r, \mathbf{b} = \mathbf{0}, \delta = 0. \quad (17)$$

The following procedure handles both random and deterministic outcomes uniformly.

Step 1 — Commutation Vector. Compute the symplectic inner product of P against each stabilizer generator:

$$c_k = \mathbf{a} \cdot \mathbf{x}_k - \mathbf{b} \cdot \mathbf{z}_k \pmod{d}. \quad (18)$$

Set $\eta = \gcd(d, c_1, \dots, c_\ell)$.

Step 2 — Isolate via SNF. Compute the Smith Normal Form of the row vector $C = [c_1 \dots c_\ell] \pmod{d}$, yielding a unimodular V such that

$$CV = [\eta \ 0 \ \dots \ 0]. \quad (19)$$

Apply V to the full tableau using the unimodular column transformation. After this change of basis, only column 1 has a nonzero commutation value η . Set $s = d/\eta$.

Step 3 — Eigenvalue of P^s . The operator P^s commutes with the entire stabilizer group (since $s \cdot \eta \equiv 0 \pmod{d}$), so the stabilizer already pins down its eigenvalue. Compute P^s via the power rule:

$$\varphi_{P^s} = s\delta + s(s-1)(\mathbf{a} \cdot \mathbf{b}) \quad (20)$$

$$\mathbf{z}_{P^s} = s\mathbf{a}, \quad \mathbf{x}_{P^s} = s\mathbf{b}. \quad (21)$$

To find the eigenvalue, decompose P^s over the generators by solving $G\mathbf{c} \equiv \mathbf{v}_{P^s} \pmod{d}$, where G is the $2n \times \ell$ Weyl block and $\mathbf{v}_{P^s} = \mathbf{z}_{P^s} \oplus \mathbf{x}_{P^s}$. The SNF of G yields $UGV = D$, reducing to a diagonal system $D\mathbf{y} \equiv U\mathbf{v}_{P^s} \pmod{d}$ with $\mathbf{c} = V\mathbf{y}$. Record the first invariant factor $f_1 = D_{11}$.

Evaluate the ordered product $Q = \prod_k S_k^{c_k}$ using the product and power rules. The Weyl part of Q equals P^s ; the phase difference $t = \varphi_Q - \varphi_{P^s} \pmod{D}$ encodes the eigenvalue.

Step 4 — Sample the Outcome. The measurement outcome h satisfies

$$2sh \equiv t \pmod{D}. \quad (22)$$

Sample h uniformly from the solution set $h_0 + \eta\mathbb{Z}_{d/\eta}$, which has $d/\eta = s$ equally likely outcomes. When $\eta = d$ (so $s = 1$), there is exactly one solution (deterministic).

Step 5 — Collapse the Tableau. Update the tableau to reflect the post-measurement state. Scale column 1 by s (via generator scaling), which zeroes out its commutation with P , and insert the measurement result $R = \tau^{-(\delta+2h)} Z^a X^b$ as a new generator.

If $S_1^s = 1$ (the scaled column is all zeros mod d), drop it. Otherwise, check whether $f_1 \mid s$. If so, S_1^s is already in the span of the other generators and can be dropped; if not, keep it (this increases ℓ by 1, up to a maximum of $2n$).

When $\eta = d$, column 1 already commutes with P , the scaling is trivial ($s = 1$), and P is already in the stabilizer group. The tableau is unchanged.

IV. REFERENCES

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V. APPENDIX: SYNDROME EXTRACTION CIRCUIT

Alternatively, $P = \bigotimes_j X_j^{a_j} Z_j^{b_j}$ can be measured indirectly via an ancilla r in $|0\rangle$ using the circuit

$$\Lambda P = (S_r^{-\mathbf{a} \cdot \mathbf{b}} \otimes 1) \prod_j \text{CZ}_{r,j}^{a_j} \prod_j \text{SUM}_{r \rightarrow j}^{b_j} \quad (23)$$

sandwiched by $F_r, F_r^\dagger$ on the ancilla, followed by measuring Z_r . The direct protocol above avoids this decomposition.